MILITARY
SPECIFICATION

MIL-S-55160(SigC)
1 February 1962

SIMULATOR, RECOGNITION SIGNAL, SM-140( )/TPX

1. SCOPE

1.1 This specification covers one type of IFF signal simulator designated, Simulator, Recognition Signal SM-140( )/TPX. (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of inviation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form - Q.

MILITARY

MIL-V-173	Varnish, Moisture and Fungus-Resistant for the Treatment of Communication, Electronic, and Associated Electrical Equipment.
MIL-S-901	Shockproof Equipment, Class H-1 (High Impact), Shipload.
MIL-C-3098	Crystal Units, Quartz.
MIL-W-6858	Welding: Aluminum, Magnesium, Nonhardening Steels or Alloys, and Titanium, Spot, Seam and Stitch.
MIL-M-10578	Metal Conditioner and Rust Remover.
MIL-I-11748	Interference Reduction for Electronic Equipment.
MIL-P-11809	Packaging and Packing of Radar Identification Sets AN/TPX-19 and An/TPX-20.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)

STANDARDS

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MIL-STD-22	Welded-Joint Designs

FSC 5895

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MIL-STD-105	Sampling for Expensive Testing by Attributes
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-B-61578	Grounding to Chassis
SC-DL-157810	Simulator, Recognition, Signal, SM-140( )/TPX.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. - Simulator, Recognition Signal, SM-140( )/TPX provides a coding signal of set characteristics to be used for an operational check of certain IFF systems.

3.2 Preproduction samples. - The contractor shall furnish two (2) preproduction samples for approval if required by the invitation for bids and the contract. (See 4.3)

3.3 Construction. - Physical construction of the item on contract shall be in accordance with the drawings and documents listed in Signal Corps drawing and data list SC-DL-157810 and in accordance with the following paragraphs.

3.3.1 Cleaning. -

3.3.1.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted.

3.3.1.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.3.2 Finish. - The equipment shall be finished in accordance with specification MIL-F-14072 and the equipment drawings. (See 4.4).

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3.3.3 Marking. -

3.3.3.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be group 1 in accordance with Specification MIL-M-13231. (See 4.4)

3.3.3.2 Visibility. - Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.3.3.3 Serial numbers. - Simulator, Recognition Signal SM-140( )/TPX shall be serial numbered.

3.3.4 Riveting. - Riveting shall not be used for mounting parts such as capacitors, resistors, transformers, inductors and tube sockets. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.3.5 Soldering. -

3.3.5.1 Solder. - Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes; and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.3.5.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.3.5.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.6 Tropicalization of material. - Material shall be treated as follows:

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(a) Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

(b) Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

(c) Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

(d) Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.1 (h).)

3.3.7 Welding. - Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

3.3.7.1 Cleaning prior to welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.3.7.2 Process. - Welds shall have thorough penetration and good fusion and shall be free from seals, blisters, abnormal pock marks, cracks, voids, slag inclusion, and other harmful defects.

3.3.7.3 Cleaning after welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat pessivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.3.8 Connections. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall

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be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Electrical connections shall not be made by clamping between a metallic and a nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to specification MIL-V-173. No varnish, lacquer, inspection paint or other coating shall be applied to completed electrical connections.

3.3.8.1 Grounding to chassis. - Grounding to the chassis shall conform to Drawing SC-B-61578.

3.4 Interchangeability. - Like units, assemblies subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the equipment. (See 4.6). Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.5 Service conditions. - The equipment shall meet the following service conditions -- where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.5.1 Operation. - Continuous use for a period of one year under the worldwide environmental conditions specified in 3.5.2 through 3.5.8, with a duty cycle of 23 hours and one hour off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc. he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with the requirement.

3.5.2 Temperature (see 4.7.1). -

(a) Operating: Ambient temperature in the range of +131°F to -40°F. (The 131°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F: exposure at the high temperature extreme not to exceed 24 hours. at anv one time.

3.5.3 Relative humidity (see 4.7.2). - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with consideration. for 4 hours.

3.5.4 Elevation (see 4.7.3). -

(a) Operating: Up to 10,000 feet above sea level.

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(b) Nonoperating: Up to 50,000 feet above sea level.

3.5.6 Vibration, internal. - The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.7.4.)

3.5.7 Bounce. - The equipment shall pass the tests of 4.9.5.1 and 4.9.6.3, and show no physical damage, after subjection to the bounce conditions of 4.7.5.

3.5.8 Shock requirements. - The equipment shall pass the tests of 4.9.5.1 and 4.9.6.3 after subjection to the bench handling and ballistic shock conditions of 4.7.7 and 4.7.6. There shall be no physical damage after the bench handling test, and only minor physical damage after ballistic shocks.

3.6 Suppresion of radio interference. - The equipment shall conform to Specification MIL-I-11748A for class 1a.

3.7 Preconditioning. - Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.8. (Also see 4.5.)

3.8 Electrical requirements. -

3.8.1 Primary power input. - The signal simulator shall operate as specified herin when operated from a primary power source of 105 to 125 volts ac, 60 or 400 cycles per second, single phase.

3.8.2 Trigger input. - The operation of the signal simulator shall be as specified herein with trigger input as follows:

(a) Trigger amplitude: 12.5 volts minimum

(b) Trigger duration: 0.5 to 1.0 microseconds

(c) Double pulse spacing: 5.0 ±.2 microseconds with TRIGGER IN switch in MODE 2 position

(d) Single pulse input: Single pulses as described in (a) and (b) above when TRIGGER IN switch is in PULSE position.

3.8.2.1 Single pulse rejection. - When the TRIGGER IN switch is in the MODE 2 position, the signal simulator shall not be triggered by single pulse input pulses of 50 volts or less.

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3.8.3 Trigger delay. - The input trigger shall be delayed a minimum of 100 usec.

Note: Trigger delay is defined as the time between the leading edge of the second input trigger of a pulse pair (or leading edge of the input trigger of a single trigger) and the leading edge of the first pulse of an output pulse train.

3.8.4 Encoder operation. -

3.8.4.1 Pulse train characteristics. - When the OUTPUT switch is in the CODE position, the output of the encoder section shall consist of a train of eight pulses with the following characteristics:

(a) Interpulse spacing. - 2.9 ±0.1 microseconds, from leading edge to leading edge, adjustable by means of the SPACING control. However, the total pulse train length from the leading edge of the first pulse to the leading edge of the eighth pulse shall be between 19.8 and 20.8 usec.

(b) Pulse duration. - 0.8 ±0.1 microseconds at 50 percent of maximum amplitude.

(c) Pulse train droop. - Not more than 5 percent of the amplitude of the highest pulse in the train.

(d) Pulse rise and decay time. - Shall rise to maximum within 0.1 usec and decay to zero within 0.2 usec.

3.8.4.2 Single pulse output characteristics. - When the OUTPUT switch is in the PULSE position, the output of the encoder section shall consist of a single pulse 0.8 ±0.1 microsecond in duration.

3.8.5 RF Section operation. -

3.8.5.1 Crystal frequency. - The crystal unit installed in the set shall operate at a resonant frequency of 40.37037 ±.00741 megacycles (1/27 of 1090 megacycles ±200 kilocycles) unless otherwise specified in the contract. In any case the crystal oscillator frequency shall be 1/27 of the desired RF output frequency.

3.8.5.2 Output frequency range. - The RF section shall be capable of being tuned over the frequency range of 1080 megacycles to 1130 megacycles. After final inspection the frequency shall be set at 1090 megacycles, ±200 kilocycles.

3.8.5.3 RF Power output. - The peak power output shall be greater than -30 dbm when operated into a 50 ohm load.

3.8.5.4 Output pulse characteristics. - The RF output shall consist of pulsed RF signals, with a pulse duration of 0.8 ±0.1 microseconds, and shall appear as single pulses, or eight-pulse trains as described in 3.8.4.1.

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3.8.6 B + control relay operation. - Upon application of 70 volts dc to the solenoid of the B + control relay, the relay shall operate to apply B+ voltage to the various signal simulator circuits.

3.8.7 System operation. - The signal simulator shall operate in conjunction with a Mark X (SIF) system, consisting of Receiver-Transmitter, Radio, RT-211A/TPX, Coder-Control, Interrogator Set, KY-97B/TPX, and Decoder Group AN/TPA-3, to provide test signals and to simulate a Code 77 reply signal when triggered by a Mode 2 interrogation signal, and shall perform other functions as required to meet the system operational tests of 4.9.8.

3.9 Government loaned property. - The following items, required for certain tests in section 4, will be loaned to the contractor for the duration of the contract:

(a) Pulse Generator AN/UPM-15
(b) Receiver-Transmitter, Radio RT-211A/TPX
(c) Crystal Unit CR-23/U, per Specification MIL-C-3098, for the following frequencies:

Operating frequency	Crystal frequency
200	.00741 Mc
1080 Mc ±0 Kc	40.00000 ± 0
0	0
1130 Mc ±200 Kc	41.85185 ±.00741 Mc

(d) Coder-Control, Interrogator Set KY-97B/TPX
(e) Decoder Group AN/TPA-3

Note: Items (d) and (e) are required for the operational test of 4.9.8 only.

3.10 Technical literature, and running spare parts. - Technical literature, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.1(f) and (g).)

3.11 Workmanship. - The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3.1 Cleaning	3.3.6 Tropicalization
3.3.4 Riveting	3.3.7 Welding
3.3.5 Soldering	3.3.8 Connections

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. - Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to

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submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery.) (See 4.3)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4. and 4.5)

(2) Acceptance inspection of preparation for delivery (See 4.11)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary document covering the items listed in 4.4, and the inspection specified for group A, group B and group C, (see tables II, III, and IV respectively). The preproduction inspection will normally be performed in this order: (a) vibration, (b) bounce, (c) shock, benchhandling, and (d) shock, ballistic; other preproduction inspection may precede, follow or be interspersed between the foregoing.

Table I. - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req. Para.	Insp. Para.
Service conditions	3.5	4.7

4.4 Acceptance Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Finish	3.3.2	Suppression of RF interference	3.6
Marking	3.3.3		

4.5 Acceptance inspection of equipment before preparation for delivery. - The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4. through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to

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control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the inspection specified herein, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, group C inspection shall be preconditioned after final assembly. (See 3.7.)

4.5.1 Group A inspection. - This inspection (including sampling) shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operation inspection (4.9.8) shall be last.

Table II. - Group A inspection

Inspection	Req. Para.	Insp Para.	AQL
Visual and mechanical, major	3.11	4.10	1.5 dphu
Visual and mechanical, minor	3.11	4.10	6.5 dphu
Electrical tests	3.8	4.9	
Trigger input	3.8.2	4.9.3.	)
Trigger characteristics	3.8.2	4.9.3.1	)
Single pulse input	3.8.2(d)	4.9.3.2	)
Single pulse rejection	3.8.2.1	4.9.3.3	) 2.5 dphu
Trigger delay	3.8.3	4.9.4	) for the
Encoder operation	3.8.4	4.9.5	) group
Interpulse spacing	3.8.4.1(a)	4.9.5.1	)
Pulse duration	3.8.4.1(b)	4.9.5.2	)
Pulse train droop	3.8.4.1(c)	4.9.5.3	)
Single pulse output characteristics	3.8.4.2	4.9.5.4	)
RF Operation	3.8.5	4.9.6	)
RF output frequency	3.8.5.2	4.9.6.2	)
RF power output	3.8.5.3	4.9.6.3	)
Output pulse characteristics	3.8.5.4	4.9.6.4	)
System operation	3.8.7	4.9.8	1%

dphu = defects per hundred units

4.5.2 Group B inspection. - This inspection (including sampling) shall conform to table IV and to the special procedures for small-sample inspection of Standard MIL-STD-105. The inspection level for normal and tightened inspection shall be L-7 and for reduced inspection shall be L-5. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

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Table III. - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL (See 4.5.2)
Crystal frequency	3.8.5.1	4.9.6.1	)
Output frequency range	3.8.5.2	4.9.6.2	) 6.5%
B + Control relay test	3.8.6	4.9.7	) for
Primary power input	3.8.1	4.9.2	) group
Interchangeability	3.4	4.6	6.5%

4.5.2.1 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection. - This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV. - Group C inspection

Inspection	Req. Para.	Insp. Para.
Interference suppression	3.6	4.4

4.5.3.1 Sampling for inspection of equipment. - Two units for each group C inspection shall be selected from each 200 units produced, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

4.5.3.2 Noncompliance. - If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4.)

4.5.4 Reinspection of conforming group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

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4.6 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.4. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of the case, cover and the insertable assembly.

(b) Location of mounting facilities which effect mating of the case or the mounting plate and mating of the mounting plate to Receiver-Transmitter, Radio RT-211( )/TPx.

4.7 Service conditions. -

4.7.1 Temperature test. - The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. The following measurements shall be taken at steps 1, 4, 8, and 10;

Paragraph	Measurement	Test results
4.9.5.1	Interpulse spacing	Specified performance
4.9.6.3	RF power output	Specified performance

4.7.2 Moisture-resistance test for equipment. - The equipment shall be tested as follwos:

4.7.2.1 Test conditions. -

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.7.2.2 Test method. -

(a) Dry at 130°±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.7.2.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.7.2.3 shall be made during the periods shown on the standard and shall comply with 4.7.2.3.

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(e) After cycling has been completed, condition the equipment for 24 hours at 77°±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.7.2.3.

4.7.2.3 Performance. - Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	Measurement	Performance during cycling
4.9.5.1	Interpulse spacing	specified performance
4.9.6.3	RF power output	Specified performance

4.7.2.4 Failure. - If any equipment fails to meet the performance specified in 4.7.2.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.7.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.7.2.2(e) it does not pass the test.

4.7.3 Elevation test. - The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.9.5.1	Interpulse spacing
4.9.6.3	RF power output

(b) Reduce the barometric pressure to 20.6±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4±0,1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for these measurements.

4.7.4 Test for internal vibration. - Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.5.6.

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(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.7.5 Bounce test. - The equipment shall be tested on the package tester, as made by the L. A. B. Corporation, Skaneteles, New York, or equal, as follows:

(a) Secure the equipment to the L. A. B., 48-inch, 305-pound, vehicular adapter plate, or equal, in the same manner as it would be mounted on Receiver-Transmitter, Radio RT-211( )/TPX. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(c) Place the adapter plate on the bed of the package tester, limiting the lateral motion (by wooden fences) to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the back-stop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of 3 hours at 283±1rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction.

4.7.6 Shock test; ballistic. - The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The equipment shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to

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reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.7.7 Shock test; bench-handling. - The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch firbench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.8 Bounce preconditioning. - The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.9 Electrical tests. - Except where otherwise specified the electrical tests shall be performed with a nominal power input of 115 volts, 60 cycles per second.

4.9.1 Test equipment required. -

(a) Input trigger source. - Adjustable single pulse source, 0.5 to 1.0 microsecond duration at 50% maximum amplitude, and paired pulses 5.0 usec, adjustable ±0.2 usec. The signal source pulse pair spacing shall be calibrated to within ± 1 percent, and the amplitude and duration to within ±5 percent.

(b) Indicator. - Oscilloscope, Hewlett-Packard 150A with Hewlett-Packard 152B Dual Channel Amplifier, or equal.

(c) RF signal source. - Signal Generator, SG-97( )/FRC (Hewlett-Packard 614A) or equal.

(d) Monitor. - Receiver-Transmitter, Radio RT-211A/TPX. (The attenuation of the RF probe shall be calibrated to within ±1 db at 1090 megacycles.)

4.9.2 Primary power input tests. - The tests of 4.9.3 through 4.9.6.3 shall be performed with primary power inputs of 105 volts, 60 cycles per second; 125 volts,

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60 cycles per second; 105 volts, 400 cycles per second; and 125 volts, 400 cycles per second to determine compliance with 3.8.1.

4.9.3 Trigger input tests. - The tests of 4.9.3.1 through 4.9.5 shall be performed using the test setup of figure 1.

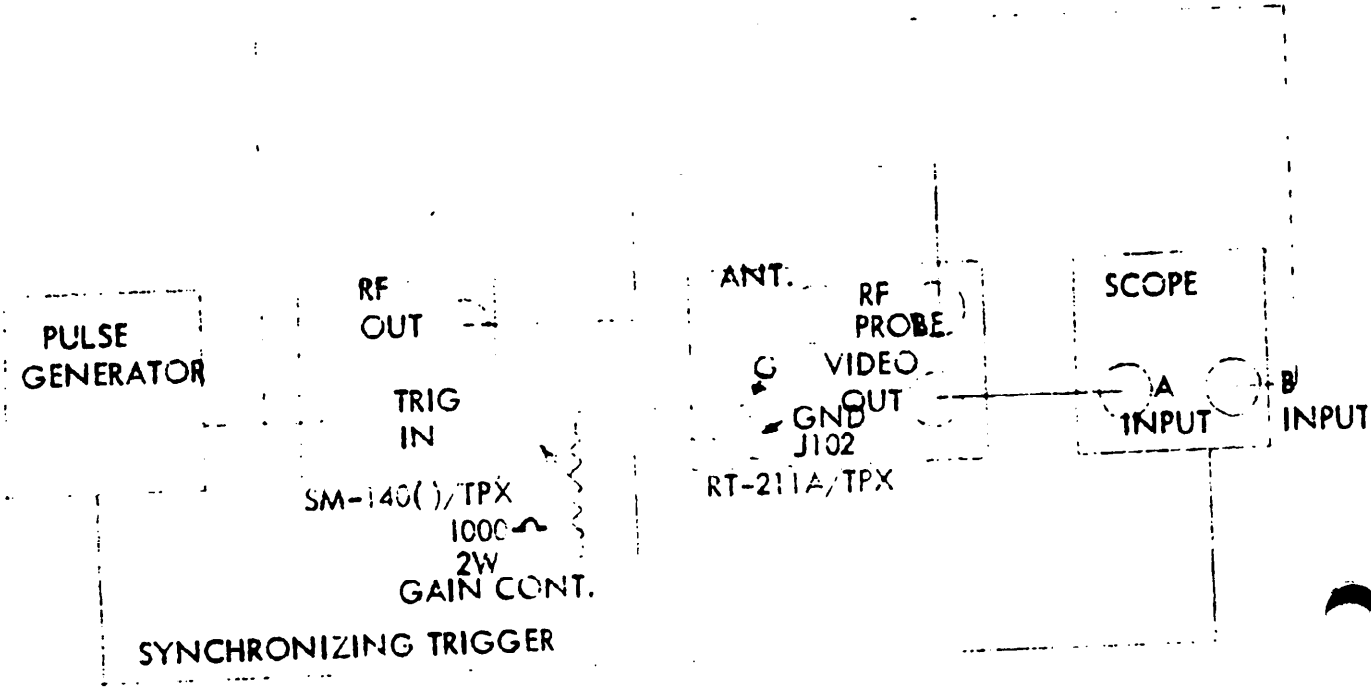


Figure I - Test Set Up

4.9.3.1 Trigger characteristics tests. - The pulse generator shall be adjusted to produce paired output pulses spaced 5.0 ±.2 microseconds apart at a pulse repetition frequency of approximately 200 pulses per second and the TRIGGER in switch of the simulator placed in the MODE 2 position. The input trigger pulse duration shall be adjusted for 0.75 microsecond pulses, with an amplitude of approximately 15 volts. The receiver gain at the RT-211A/TPX shall be adjusted to produce a non-limited video presentation on the oscilloscope. The following tests shall be performed:

(a) The output signal shall be observed and its character noted for reference.

(b) The amplitude of the input trigger shall be reduced to 12.5 volts and the output observed.

(c) Repeat (b) with a ten volt trigger input.

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(d) The duration of the input triggers shall be reduced to 0.5 microseconds and increased to 1.0 microseconds and the output signal observed in each case.

(e) The input trigger pulse spacing shall be increased to 5.2 microseconds and decreased to 4.8 microseconds and the output signal observed in each case.

(f) The output signal observed in steps (b) through (e) shall be the same as the signal observed in step (a).

4.9.3.2 Single pulse input test. - The pulse generator shall be adjusted to produce a single pulse output of 0.75 microseconds duration and 15 volts amplitude. The TRIGGER IN switch of the simulator shall be placed in the PULSE position and the video output signal of the RT-211A/TPX observed. The output shall not have changed from that observed in 4.9.3.1(a).

4.9.3.3 Single pulse rejection test. - The TRIGGER IN switch shall be placed in the MODE 2 position and the amplitude of the single pulse output of the pulse generator increased, to 50 volts. The video output shall be observed to determine compliance with 3.8.2.1.

4.9.4 Trigger delay test. - The time between the input trigger and first pulse of the output pulse train shall be measured by feeding the input triggers into one channel, and the video output of the RT-211A/TPX into the other channel of the dual channel amplifier of the oscilloscope as indicated in figure 1. The DELAY control of the simulator shall be adjusted to the minimum and maximum positions and the delay measured to determine compliance with 3.8.3.

4.9.5 Encoder operation test. - This test shall be performed with the equipment connected as in Figure 1 and the simulator triggered by paired pulses.

4.9.5.1 Interpulse spacing. - The OUTPUT switch shall be placed in the CODE position and the time between the leading edges of any two consecutive pulses measured to determine compliance with 3.8.4.1(a). The total pulse train length (8 pulses) shall also be measured for compliance with the pulse train length requirement of 3.8.4.1(a).

4.9.5.2 Pulse duration. - The time between the 50 percent amplitude point on the leading edge and the 50 percent point on the trailing edge of a pulse shall be measured, and shall be as specified in 3.8.4.1(b).

4.9.5.3 Pulse train droop. - The amplitudes of the highest pulse (A_1) and lowest pulse (A_2) of the train shall be measured and the percentage of pulse train droop calculated using the formula: % droop = $\frac{A_1 - A_2}{A_1}$ (100). The results of this test shall confirm compliance with 3.8.4.1(c).

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4.9.5.4 Single pulse output characteristics. - The OUTPUT switch shall be placed in the PULSE position and the oscilloscope presentation observed to determine that a single pulse is present. The pulse duration shall be measured as in 4.9.5.2 to determine compliance with 3.8.4.2.

4.9.6 RF section tests. -

4.9.6.1 Crystal frequency. - The operating frequency of the crystal unit shall be measured with the crystal in the simulator. The frequency meter used for this test shall have an accuracy of at least .001 percent.

4.9.6.2 Output frequency range. - Crystal units for operating frequencies of 1080 megacycles and 1130 megacycles (crystal frequencies of 40.00000 megacycles and 41.85185 megacycles respectively) shall be inserted in the set and the RF section tuned to these frequencies. The output RF frequency shall be measured at each frequency to determine compliance with 3.8.5.2.

4.9.6.3 RF power output. - The RF power output shall be measured at an RF output frequency 1090 megacycles ±200 kilocycles as follows:

(a) The test set-up of figure 1 shall be used, with the addition of a signal generator (See 4.9.1(c)) fed into the antenna terminal of the RT-211A/TPX. Adjust the signal generator to produce 1.0 microsecond pulses at the simulator operating frequency.

(b) The OUTPUT LEVEL control of the simulator shall be adjusted to maximum and the RT-211A/TPX receiver gain adjusted to produce an unlimited signal presentation on the oscilloscope.

(c) The position of the signal from the signal generator shall be adjusted to place it adjacent to one of the simulator signals and the amplitude adjusted until the signal generator output observed on the scope is the same amplitude as the simulator output signal.

(d) The signal generator power output shall be determined and the simulator power output calculated, taking into account the attenuation of the RF probe of the RT-211A/TPX and the differences in cable attenuation.

(e) The OUTPUT LEVEL control shall then be adjusted to the minimum position and the power output again measured in the same manner.

(f) The power output, as measured in this test, shall be as specified in 3.8.5.3.

4.9.6.4 RF output pulse characteristics. - The characteristics of the RF output pulses shall be measured by observing the video output of the RT-211A/TPX (see figure 1), and shall conform to the requirements of 3.8.5.4.

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4.9.7 B + control relay test. - The operation of the B + control relay shall be tested by applying 70 volts dc to the relay solenoid while monitoring the B+ voltage in the set. The operation of the relay shall be as specified in 3.8.6.

4.9.8 System operational test. - The simulator shall be connected into a Mark X(SIF) system, consisting of Receiver-Transmitter, Radio RT-211A/TPX, Coder-Control, Interrogator Set KY-97B/TPX and Decoder Group AN/TPA-3 together with the necessary interconnecting cables. Any appropriate oscilloscope may be used for an indicator. The following operational checks shall be performed:

(a) The KY-97B/TPX shall be operated in MODE 2 and the AN/TPA-3 in Code 77. The simulator shall be placed in operation with the TRIGGER IN switch in the MODE 2 position and the OUTPUT switch in the CODE position. The video signal from the AN/TPA-3 and RT-211A/TPX shall be observed, and shall consist of a single pulse from the AN/TPA-3, and an eight-pulse train from the RT-211A/TPX.

(b) The simulator shall be operated as in step (a) while the KY-97B/TPX is operated in MODE 1 and MODE 3. There shall be no output from the AN/TPA-3 or from the video output of the RT-211A/TPX.

(c) The TRIGGER IN switch of the simulator shall be placed in the PULSE position and the KY-97B/TPX operated to produce a single pulse output (by removing one of the pulse output tubes). There shall be a single pulse output from the AN/TPA-3.

(d) The OUTPUT switch of the simulator shall be placed in the PULSE position and the video output of the RT-211A/TPX observed to determine that a single pulse is produced.

(e) The output shall be observed while the CHOP switch of the KY-97B/TPX is operated on and off with the B+ ON switch of the simulator in the REMOTE position. The output shall be present when the CHOP switch is on and absent when the CHOP switch is off.

4.10 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table V:

Table V. - Classification of visual and mechanical defects.

Classification	Defect
Major	Cover insulating gasket missing
	Improper fit of cover
	Improper fit of cover gasket
	Cover or case bent or warped.
	Control inoperative
	Improper dressing of RF cable shielding

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Table V. - Classification of visual and mechanical defects. (Continued)

Classification	Defect
Minor	Cover screw(s) missing Control difficult to operate Difficulty in obtaining good contact or in making the required setting of a control due to play or backlash.

4.11 Acceptance inspection of preparation for delivery. - Acceptance inspection of preparation for delivery shall be in accordance with the applicable provisions of Specification MIL-P-11809.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging shall be in accordance with the applicable paragraphs of Specification MIL-P-11809.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) Preproduction inspection:

(1) See 3.2.

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).
(b) Number of each kind of pack to be submitted.
(c) Inspection to be performed thereon.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.10)

(g) Quantity of running spare parts required. (See 3.10)

(h) Submission of the statement of treatment referenced in 3.3.6(d) as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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6.2 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example SM-140W/TPX. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.3 Definition of level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.4 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL-STD-105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced provisions of Standard MIL-STD-105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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